FIITJEE **JEE (Main), 2018**

ALL INDIA TEST SERIES

FULL TEST – I

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

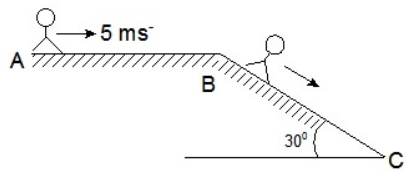
Physics

PART – I

SECTION – A
Straight Objective Type

This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. A man moving on track ABC, getting rain drops vertical for the path AB, and along same line of its motion for path BC then actual speed of rain drop is



(A) $10\sqrt{3}ms^{-1}$

(B) $\frac{10}{\sqrt{3}}ms^{-1}$

(C) $5\sqrt{3}ms^{-1}$

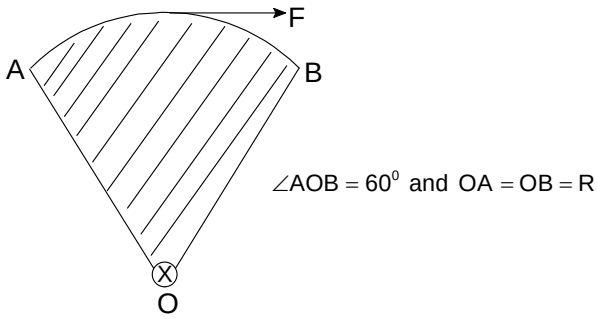
(D) Can't find

2. Two bodies A and B are at positions (2, 3) m and (5, 7) m respectively. These two are moving with constant velocity 3 ms^{-1} along –ve y axis and 4ms^{-1} along +ve x axis respectively. Minimum separation between them during their motion is

(A) 5 m (B) 3 m

(C) 4 m (D) 10 m

3. A force F acts tangentially on a part of a uniform disc of mass 'm' and radius 'R' hinged at point O. The force exerted by hinge on the portion of disc initially.



(A) $\left(\frac{4-\pi}{\pi}\right)F$

(B) $\frac{4F}{\pi}$

(C) $F\left(\frac{\pi-2}{\pi}\right)$

(D) $\frac{F}{2}$

Space for rough work

4. If acceleration gradient of a moving body along positive x-axis is $-4s^{-2}$ and body starts its motion from rest from x = 5m, having maximum acceleration (magnitude) then velocity of moving body at x = -3m is

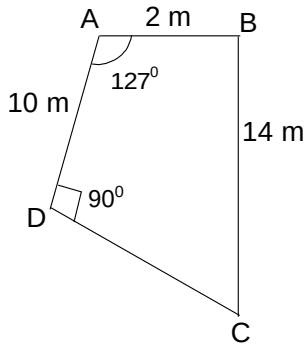
(A) 6 ms^{-1} (B) 4 ms^{-1}

(C) 8 ms^{-1} (D) Cant reach at x = -3m

5. In the figure shown ABCD is a lamina of mass 12 kg. The moment of inertia of the given body about an axis passing through D, and perpendicular to the plane of the body is

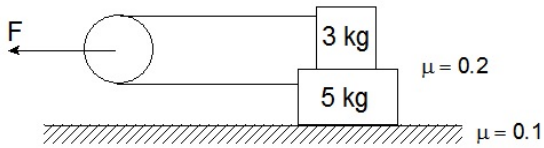
(A) 256 kg m^2 (B) 512 kg m^2

(C) 200 kg m^2 (D) 100 kg m^2



6. For the arrangement as shown in the figure, the maximum force F, for no relative motion between blocks is

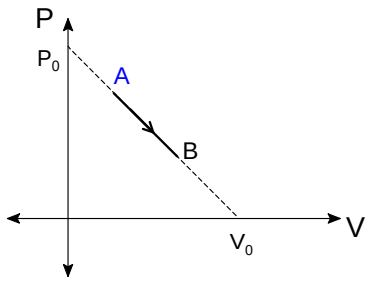
(A) 12 N
(B) 8 N
(C) 20 N
(D) 24 N



7. One mole of a monoatomic gas undergoes a linear process $A \rightarrow B$ in P – v diagram. The volume of the gas when the process turns from an endothermic to an exothermic one is

(A) $\frac{5}{8}v_0$ (B) $\frac{v_0}{2}$

(C) $\frac{8v_0}{15}$ (D) $\frac{3v_0}{5}$



Space for rough work

8. Two coaxial solenoids having same number of turns per unit length n, and radius R and 2R respectively have no current initially. Current in both grows uniformly such that current in the inner one is twice that of the outer one, at any instant of time in the same direction. Current in outer one at time t is Kt then induced electric field at a distance r from axis is (R < r < 2R).

(A) Not possible

(B) $\frac{\mu_0 nK}{2r}\left(2R^2 + r^2\right)$

(C) $\frac{2\mu_0 nK.R^2}{r}$

(D) $\frac{2\mu_0 nK}{r}\left(2R^2 - r^2\right)$

9. A particle of mass m moves in a force field such that its potential energy in force field is defined by the equation $U = +A(x-a)^2(x-b)^2$. Where A, a and b are +ve constants then body may oscillate simple harmonically about point.

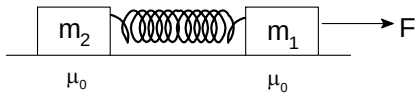
(A) a only

(B) a and b both

(C) only $\frac{a+b}{2}$

(D) b only

10. Find minimum value of F such that m_2 starts its motion on the ground



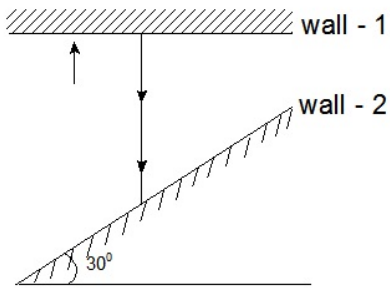
(A) $\frac{\mu_0(2m_1 + m_2)g}{2}$

(B) $\mu_0(2m_1 + m_2)g$

(C) $\mu_0(m_1 + m_2)g$

(D) $\frac{\mu_0(m_1 + m_2)g}{2}$

11. Molecule of a gas in a container hits one wall (1) normally and rebounds back. It suffers number of collision and hits the opposite wall (2) which is at angle of 30^0 with wall 1. Assuming the collision to be elastic and the small collision time to be the same for both the walls. The magnitude of average force by walls and 2 are F_1 and F_2 provide to molecule during collision satisfy



(A) $F_1 > F_2$

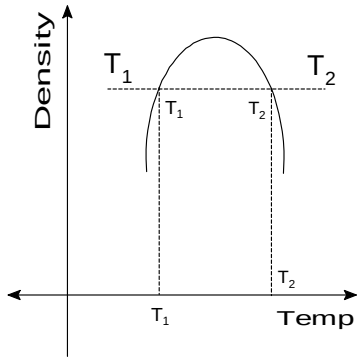
(B) $F_1 < F_2$

(C) $F_1 = F_2$ both non zero

(D) $F_{12}F_{220}$

Space for rough work

12. A thermometer uses density of water as thermometric property. Actual reading in thermometer is height of water (h) which inversely proportional to the density of water (d). In certain temp range density of water varies with temp as shown in the graph is symmetrical about maximum.
Two identical bodies (of same mass and specific heat) at different temp T_1 and T_2 show the same reading of height $h_1 = h_2$ in the thermometer. The bodies are brought in contact and allowed to reach the equilibrium. The thermometric reading height of water for final equilibrium state h_f satisfies



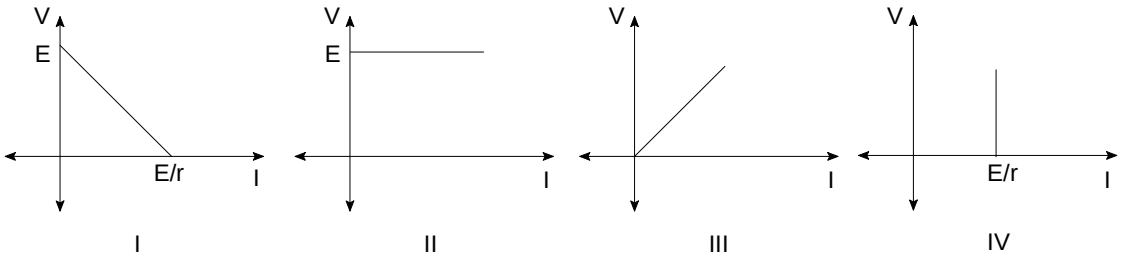
(A) $h_f = \frac{h_1 + h_2}{2} = h_1 = h_2$

(B) $h_f > h_1 = h_2$

(C) $h_f, h_1 = h_2$

(D) h_f may be greater or less than $h_1 - h_2$

13. A power supply of fixed emf E is connected to a variable resistor (resistance = R) and an ammeter of resistance R_A. The internal resistance of supply is r. The potential difference between the terminals of the power supply in closed circuit is denoted by V and current in circuit denoted by I. A graph is obtained between V and I by drawing varying amount of current from the supply. The correct V – I graph for the given situation is



(A) Graph I
(B) Graph II
(C) Graph III
(D) Graph IV

Space for rough work

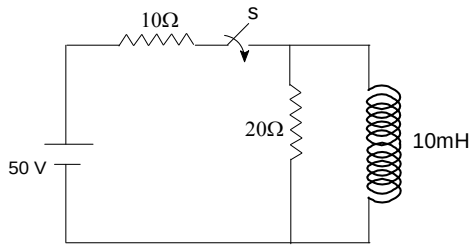
14. If the switch is closed for long time and then open at t = 0, the P.D across indicator at t = 0 is

(A) OV

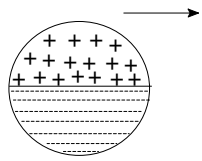
(B) 50 V

(C) 20 V

(D) 100 V



15. A disc having uniform surface charge density in upper part $+\rho$ and $-\rho$ in lower half placed on rough horizontal surface as shown in figure. A uniform electric field is set up as shown if mass of disc is M and its radius is R as well sufficient friction is present to prevent slipping. The acceleration of disc is



(A) $\frac{8}{9}\frac{\rho R^2 E_0}{M}$

(B) $\frac{9}{8}\frac{\rho R^2 E_0}{M}$

(C) $\frac{4}{3}\frac{\rho R^2 E_0}{M}$

(D) $\frac{3}{4}\frac{\rho R^2 E_0}{M}$

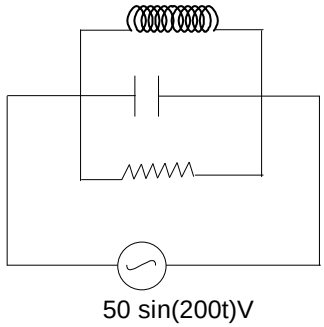
16. If L = 25 mH, C = 1 mF and R = 10Ω then R.M.S current across source is

(A) $5\sqrt{2}$ A

(B) $2.5\sqrt{2}$A

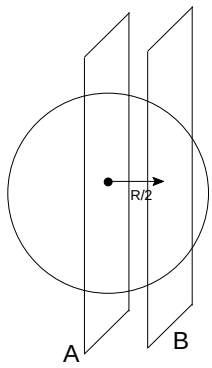
(C) 12.5 $\sqrt{2}$A

(D) $25\sqrt{2}$ A



Space for rough work

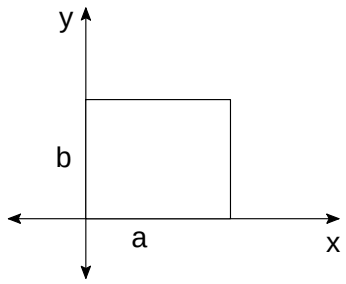
17. From a sphere of electrical conductivity K two planes cut a piece such that first plane passes through the centre of sphere and second parallel to first one at distance R/2 from centre then resistance between A and B



(A) $\frac{1}{\pi KR}$ (B) $\frac{\sqrt{3}}{\pi KR}$

(C) $\frac{\ell n3}{2\pi KR}$ (D) None

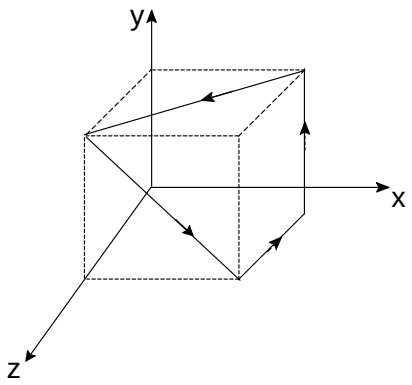
18. A thin rectangular plate with sides of length 'a' and 'b' has mass per unit area varying as $\alpha = \alpha_0\left(\frac{xy}{ab}\right)$ where α_0 is a constant. The co-ordinate of centre of mass of the plate is



(A) $\left(\frac{a}{3},\frac{b}{3}\right)$ (B) $\left(\frac{a}{2}.\frac{b}{2}\right)$

(C) $\left(\frac{4a}{3},\frac{4b}{3}\right)$ (D) $\left(\frac{2a}{3},\frac{2b}{3}\right)$

19. A wire carrying 10 A current is bent to pass through various side of a cube of side 10 cm as shown in figure. A magnetic field $\vec{B} = 2\hat{i} - 3\hat{j} + \hat{k}$ is present in region then torque on the loop.



(A) $\sqrt{20}\times10^{-1}$Nm

(B) $\sqrt{20}\times10^{-2}$Nm

(C) $\sqrt{150}\times10^{-2}$Nm

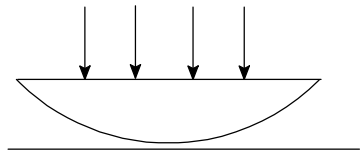
(D) $\sqrt{150}\times10^{-2}$Nm

Space for rough work

20. Above Curie temp
(A) Paramagnetic materials become ferromagnetic
(B) Ferromagnetic materials becomes diamagnetic
(C) Ferromagnetic materials becomes paramagnetic
(D) Paramagnetic materials becomes diamagnetic

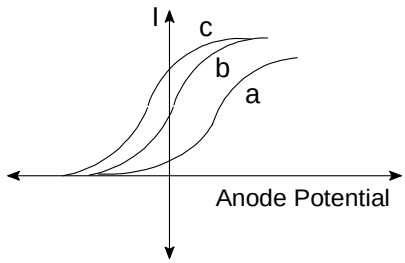
21. If a single slit fraunhoffer diffraction set up is used with light of wavelength 4000 $\overset{0}{A}$. Distance D between central maximum and first minimum found to be 0.3 cm in set up if the wavelength changed to 6000 $\overset{0}{A}$ the corresponding value of D will be
(A) 0.20 cm (B) 0.24 cm
(C) 0.30 cm (D) 0.45 cm

22. A thin slice is cut – out of a cylindrical glass along the plane parallel to its axis. The slice is placed on a flat glass plate as shown. The observed interference fringes from this combination shall be



(A) Straight
(B) Circular
(C) equally spaced
(D) having fringe spacing which increases as go out ward

23. Above figure shows variation of current with anode potential for a photo sensitive surface for there different radiation. Let I_a , I_b and I_C be intensities and f_a f_b and f_c the frequencies for curve a, b and c respectively



(A) $f_a = f_b, I_a \neq I_b$
(B) $f_a = f_c, I_a = I_c$
(C) $f_a = f_b, I_a = I_b$
(D) $f_b = f_c, I_b = I_c$

24. Lead is a good absorber of x rays. Specific gravity of Pb = 11.5. To cut down the initial radiation I_0 to $\frac{I_0}{2}$ a shield of 1.5 cm was used. The mass absorption co-efficient is
(A) 0.06 cm^2 / g (B) 0.462 cm^{-1}
(C) 0.21 cm^{-1} (D) 0.040 cm^2 / g

Space for rough work

25. M_x and M_y denote the atomic masses of the parent and daughter nuclei respectively in a radio active decay. The Q value of a β^- decay is Q_1 and for β^+ decay is Q_2. If m_e denote the masses of electron then which of following statement is correct.

(A) $Q_1 = (M_x - M_y)C^2$ $\quad Q_2 = (M_x - M_y - 2m_e)C^2$

(B) $Q_1 = (M_x - M_y)C^2$ $\quad Q_2 = (M_n - M_y)C^2$

(C) $Q_1 = (M_x - M_y - 2m_c)C^2$ $\quad Q_2 = (M_x - M_y + 2m_c)C^2$

(D) $Q_1 = (M_x - M_y + 2m_c)C^2$ $\quad Q_2 = (M_x - M_y - 2m_c)C^2$

26. Power emitted per unit area of a black body is R watt/m^2. At what wavelength will the power radiated by black body be maximum if stefan's constant is σ and Wien's constant is b then.

(A) $\lambda = \left(\frac{b\sigma}{R}\right)^{\frac{1}{4}}$ (B) $\lambda = \left(\frac{b}{R}\right)^{\frac{1}{4}}$

(C) $\left(\frac{bR}{\sigma}\right)^{\frac{1}{4}} = \lambda$ (D) $\lambda = b\left(\frac{\sigma}{R}\right)^{\frac{1}{4}}$

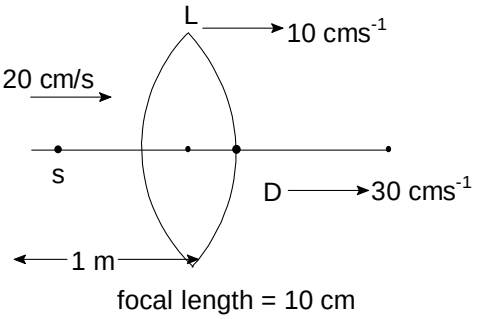
27. The figure shows the initial position of a point source of light s, a detector D and Lens L. Now at t = 0 all three starts moving toward right with different velocity as shown in figure. The times at which detector receives the maximum light.

(A) 0.56 s and 4.45 s

(B) 3.8 s

(C) 8.94 s and 19.62 s

(D) 0.56 s



Space for rough work

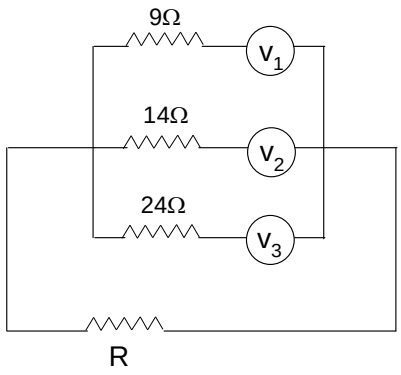
28. The figure shows a portion of a circuit. Resistor are known and indicated on diagram and voltmeter are identical. If voltmeters v_1 and v_2 reads 7.5v and 5v respectively then reading of v_3.

(A) 0v

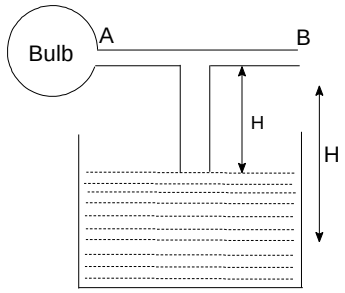
(B) 10v

(C) 2.5v

(D) 3.0v



29. A liquid of density ρ has to spray with help of aspirator pump at height H from liquid level as shown in figure. If gauge pressure inside bulb is P and density of air is σ then velocity of air in tube AB is



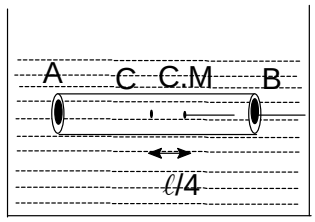
(A) $\left\{\frac{P+\rho gH}{\frac{\sigma}{2}}\right\}^{\frac{1}{2}}$

(B) $\left\{\frac{P-\rho gH}{\frac{\sigma}{2}}\right\}^{\frac{1}{2}}$

(C) $\left\{\frac{P+\alpha gH}{\frac{\sigma}{2}}\right\}^{\frac{1}{2}}$

(D) $\left\{\frac{P+\rho gH}{\sigma}\right\}^{\frac{1}{2}}$

30. A non uniform cylinder of mass m, length ℓ and radius r having its C.O.M at a distance $\frac{\ell}{4}$ from centre and lying on axis of cylinder. The cylinder is kept in a liquid of uniform density ρ. The moment of inertia of the rod about C.O.M is I. The angular acceleration of point A relative to point B just after the rod is released from position shown in figure is.



(A) $\frac{\pi\rho g\ell^2 r^2}{2I}$

(B) $\frac{3\pi\rho g\ell^2 r^2}{4I}$

(C) $\frac{\pi\rho g\ell^2 r^2}{I}$

(D) $\frac{\pi\rho g\ell^2 r^2}{4I}$

Space for rough work

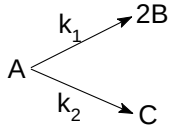
Chemistry

PART – II

SECTION – A
Straight Objective Type

This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

31. For the following elementary first order reaction:



If $k_2 = 2k_1$, then % of B in overall product is:

(A) 25% (B) 50%

(C) 75% (D) 100%

32. One of the allotropic form of boron is α – rhombohedral boron. Number of faces and corners in its basic building B_{12} icosahedral unit are:

(A) 20 and 12 respectively (B) 12 and 20 respectively

(C) 12 and 12 respectively (D) 20 and 20 respectively

33. Which of the following is bio-degradable polymer?

(A) Nylon – 6, 6 (B) Nylon – 6, 10

(C) Nylon - 6 (D) Nylon – 2-Nylon -6

34. White phosphorus is a tetra atomic solid P_4 (s) at room temperature. Find average (P – P) bond enthalpy (in kJ/mol).

Given : $\Delta H_{sublimation}$ of $P_4(s) = 59$ kJ / mol

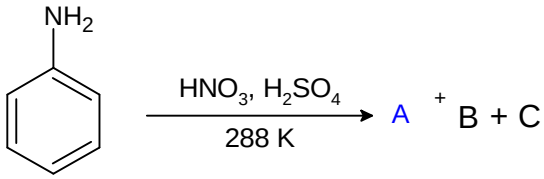
$\Delta H_{atomization}$ of $P_4(s) = 1265$ kJ / mol

(A) 201 (B) 1206

(C) 301.5 (D) 210.83

Space for rough work

35. In extraction of copper from its sulphide ore, the metal is obtained by reduction of Cu_2O with:
(A) CO (B) FeS
(C) SO_2 (D) Cu_2S

36. 

Order of basic nature is A > B > C.
Order of % of A, B and C in overall product is:
(A) A > B > C (B) C > A > B
(C) B > A > C (D) A > C > B

37. A cell contains two hydrogen electrodes. The negative electrode is in contact with a solution of pH = 5.5. The emf of the cell is 0.118 V at 25^0 C. Calculate the pH of solution at positive electrode. (assume pressure of H_2 in both electrodes = 1 bar)
(A) 3.5 (B) 7.5
(C) 4.5 (D) 6.5

38. The image formed, when a photographic film is exposed and developed, is due to formation of
(A) $\left[Ag(S_2O_3)_2\right]^{-3}$ (B) AgBr
(C) Ag (D) Ag_2O

39. Wood spirit is obtained by catalytic reaction of water gas with hydrogen at 200 – 300^0C. The catalyst used is:
(A) Ni (B) ZnO – Cr_2O_3
(C) Cu (D) Pd – $BaSO_4$

40. The compressibility factor (Z) of real gas is usually less than 1 at low temperature and low pressure because
(A) a is negligible while b is not (B) b is negligible while a is not
(C) both a and b are negligible (D) neither a nor b are negligible

Space for rough work

41. $Zn + \text{conc. } HNO_3 \rightarrow Zn(NO_3)_2 + X + H_2O$

$Zn + \text{dil. } HNO_3 \longrightarrow Zn(NO_3)_2 + Y + H_2O$

Compounds X and Y are respectively:

(A) N_2O, NO (B) NO_2, N_2O

(C) N_2O, NO_2 (D) NO_2, NO

42. Which of the following compound will give blood red colour while doing the Lassaigne's test:

(A) $H_2N-\overset{\overset{O}{\|}}{C}-NH_2$ (B) $H_2N-\overset{\overset{S}{\|}}{C}-NH_2$

(C) $C_6H_5-SO_3H$ (D) $CHCl_3$

43. On mixing which of the following solutions, observed boiling point of mixture is higher than theoretical boiling point?

(A) n – butylchloride + n - butylbromide (B) water + ethanol

(C) chloroform + diethyl ether (D) acetone + benzene

44. The electro negativity of H, X and O are 2.1, 0.8 and 3.5 respectively on Pauling scale. The nature of compound HOX is:

(A) acidic (B) basic

(C) neutral (D) amphoteric

45. Acetylene on passing through excess of HOCl solution forms

(A) ethylene chlorohydrin (B) glyoxal

(C) ethylene glycol (D) dichloroacetaldehyde

46. 50 ml, 0.1M $NaHCO_3$ aqueous solution is mixed with 50 ml, 0.8 M Na_2CO_3 aqueous solution at 25^0C. The pH of resulting solution exists between colour transition range of an indicator which is: $\left(K_{a_1} = 4\times10^{-7}, K_{a_2} = 2\times10^{-11} \text{ for } H_2CO_3\right)$

(A) Phenol red (6.8 to 8.4) (B) Propyl red (4.6 to 6.4)

(C) Phenolphthalein (8.3 to 10.1) (D) Malachite green (11.4 to 13)

Space for rough work

47. In which of the following complexes of Co, the magnitude of CFSE (Δ_0) is maximum?

(A) $\left[Co(NH_3)_6\right]^{+2}$ (B) $\left[CoCl_6\right]^{-3}$

(C) $\left[Co(C_2O_4)_3\right]^{-3}$ (D) $\left[Co(CN)_6\right]^{-3}$

48. In simplified version of nucleic acid chain, nucleotides are joined together by

(A) $\beta-$ glycosidic linkage (B) phosphodiester linkage

(C) $\alpha-$ glycosidic linkage (D) peptide linkage

49. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are:

(A) occupied by Na^+ only (B) occupied by Cl^- only

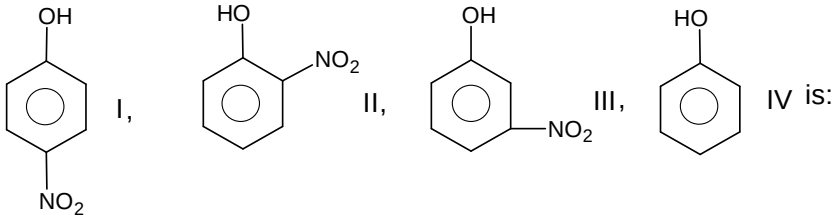
(C) occupied by either Na^+ or Cl^- (D) neither occupied by Na^+ nor Cl^-

50. Which of the following species are incorrectly matched with their shape?

(A) ClF_3 = trigonal planar (B) SF_4 = folded square

(C) $[XeF_5]^-$ = pentagonal planar (D) I_2Cl_6 = rectangular planar

51. Correct order of pK_a values of



is:

(A) I < II < III < IV (B) I > II > III > IV

(C) II < I < III < IV (D) II > I > III > IV

52. Isoelectric point of gelatin solution is 4.7. Then correct option is:

(A) If pH < 4.7, it is (-)vely charged, move towards cathode

(B) If pH < 4.7, it is (+)vely charged, move towards anode

(C) If pH > 4.7, it is (+)vely charged, move towards cathode

(D) If pH > 4.7, it is (-)vely charged, move towards anode

Space for rough work

53. $Mg(s) + C(s) \xrightarrow[\text{high T}]{\Delta} X \xrightarrow{H_2O} Mg(OH)_2 + Y$

X and Y are respectively:

(A) Mg_2C and CH_4 (B) Mg_4C_3 and CH_4

(C) Mg_2C_3 and C_3H_4 (D) MgC_2 and C_2H_2

54. In the reaction

$CF_3 - CHCl_2 \xrightarrow[\Delta]{LDA} CF_2 = CCl_2$

Intermediate formed is:

(A) carbocation (B) carbanion

(C) free radical (D) No intermediate formed

55. For an ideal diatomic gas, during any process T = KV, find out the molar heat capacity of the gas during process (neglecting vibrational degree of freedom)

(A) $\frac{3R}{2}$ (B) $\frac{5R}{2}$

(C) $\frac{7R}{2}$ (D) 2R

56. $Cu^{+2}(aq) + K_4\left[Fe(CN)_6\right](aq) \longrightarrow \underset{(ppt.)}{X}$

Colour of precipitate (X) is:

(A) green (B) blue

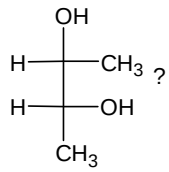
(C) white (D) reddish - brown

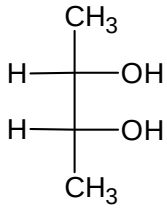
57. In which of the following set of orbitals, nodal plane is absent?

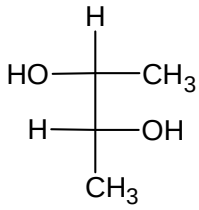
(A) $4s, 4d_{z^2}$ (B) $4s, 4d_{x^2-y^2}$

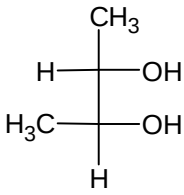
(C) $4p_z, 4s$ (D) $4d_{yz}, 4d_{zx}$

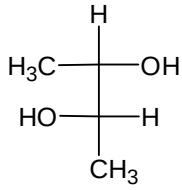
Space for rough work

58. Which of the following is enantiomer of  ?

(A) 

(B) 

(C) 

(D) 

59. Correct statement about classical smog (I) and photochemical smog (II) is:
(A) I = reducing smog, II = oxidizing smog
(B) I = oxidizing smog, II = reducing smog
(C) Both I and II are oxidizing smog
(D) Both I and II are reducing smog

60. Which of the following reaction will not proceed via carbanion intermediate?
(A) Perkin reaction
(B) Cannizaro reaction
(C) Claisen condensation
(D) Aldol condensation

Space for rough work

Mathematics

PART – III

SECTION – A
Straight Objective Type

This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

61. The maximum value of K for which $\sin x + 2x \geq kx(x+1) \ \forall \ x \in \left[0, \frac{\pi}{2}\right]$ equals

(A) $\frac{4(\pi+2)}{\pi(\pi+1)}$ (B) $\frac{4(\pi+1)}{\pi(\pi+2)}$

(C) $\frac{4\pi(\pi+1)}{\pi+2}$ (D) $\frac{4\pi(\pi+2)}{\pi+1}$

62. Differential co-efficient of $\log_{10} x$ w.r.t $\log_x 10$ is

(A) $\frac{-(\log x)^2}{(\log 10)^2}$ (B) $\frac{(\log_{10} x)^2}{(\log 10)^2}$

(C) $\frac{(\log_x 10)^2}{(\log 10)^2}$ (D) $\frac{-(\log 10)^2}{(\log x)^2}$

63. The value of $\lim\limits_{x \to \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}}$ is equal to

(A) $\frac{8}{\pi} f(2)$ (B) $\frac{2}{\pi} f(2)$

(C) $\frac{8}{\pi} f\left(\frac{1}{2}\right)$ (D) $4f(2)$

Space for rough work

64. Let $a_{n+1} = \sqrt{2 + a_n}$, n = 1, 2, 3 …. & $a_1 = 3$ then $\lim_{n\to\infty} a_n$ is

(A) π (B) 2

(C) $\sqrt{5}$ (D) 3

65. The maximum value of area of rectangle OABC, where O is origin, A lies on positive y-axis & B lies on the curve $y = \frac{\ln x}{x^2}$ is

(A) e^{-1} (B) $e^{\frac{-1}{2}}$

(C) 1 (D) $e^{\frac{1}{2}}$

66. $x > -1$, let $f(x) = \int_0^{\frac{\pi}{4}} \log_e (1 + x\tan z)\,dz$, then value of $f\left(\frac{1}{2}\right) - f\left(\frac{1}{3}\right)$ equals

(A) $\frac{\pi}{4}\log_e\left(\frac{9}{8}\right)$ (B) $\frac{\pi}{8}\log_e\left(\frac{9}{8}\right)$

(C) $\frac{\pi}{9}\log_e\left(\frac{9}{8}\right)$ (D) $\frac{\pi}{8}\log_e\left(\frac{3}{2}\right)$

67. For $t \in R$ & f be a continuous function. Let $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} x.f\left(x(2-x)\right)dx$ & $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f\left(x(2-x)\right)dx$, then $\frac{I_1}{I_2}$

(A) 1 (B) 3

(C) $\frac{1}{3}$ (D) $\frac{2}{5}$

Space for rough work

68. The general solution of the differential equation $x\left(y^2.e^{xy}+e^{\frac{x}{y}}\right)dy = y\left(e^{\frac{x}{y}}-y^2e^{xy}\right)dx$ is

(A) $e^{xy} = e^{\frac{x}{y}} + c$ (B) $e^{xy} = e^{\frac{y}{x}} + c$

(C) $e^{xy} + e^{\frac{x}{y}} = c$ (D) $e^{xy} + e^{\frac{y}{x}} = c$

69. Let z_1, z_2 & z_3 are vertices of acute angled ΔABC in anticlockwise order and origin be the circumcenter of the triangle. If bisector of $\angle A$ meets the circumcircle at point P(z), then $\arg\left(\frac{z}{z_3 - z_2}\right)$ is

(A) $\frac{\pi}{2}$ (B) $\frac{-\pi}{2}$

(C) $\frac{\pi}{6}$ (D) none of these

70. If from each of the three boxes containing 3 white & 1 black, 2 white & 2 black, 1 white & 3 black balls, one ball is drawn at random, then the probability that 2 white & 1 black ball will be drawn is

(A) $\frac{13}{32}$ (B) $\frac{1}{4}$

(C) $\frac{1}{32}$ (D) $\frac{3}{16}$

71. The set of exhaustive values of 'α' for which the equation $|\alpha x - 1| = x^2 + \alpha x + 2$ has at least one positive root, is

(A) $(-\infty, -1] \cup [1, \infty)$ (B) R

(C) $(-\infty, -1]$ (D) $[1, \infty)$

Space for rough work

72. Consider the points $A \equiv (0,1)$ & $B \equiv (2,0)$, P be a point on the line 4x + 3y + 9 = 0. Co-ordinates of the point 'P' such that |PA – PB| is maximum is

(A) $\left(\frac{-12}{5},\frac{17}{5}\right)$ (B) $\left(\frac{-84}{5},\frac{13}{5}\right)$

(C) $\left(\frac{-6}{5},\frac{17}{5}\right)$ (D) $\left(\frac{-24}{5},\frac{17}{5}\right)$

73. If the tangent at the point P(2, 4) to the parabola $y^2 = 8x$ meets the parabola $y^2 = 8x + 5$ at Q & R, then the mid point of QR is

(A) (2, 4) (B) (4, 2)

(C) (7, 9) (D) none of these

74. For an ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, θ_1 & θ_2 are the eccentric angle of points P & Q respectively, if $|\theta_1|+|\theta_2|=\pi$ then

(A) point P & Q are in opposite quadrant
(B) line PQ is parallel to major axis
(C) line PQ always passes through centre of ellipse
(D) none of these

75. The locus of point of intersection of perpendicular tangents of the hyperbola $4y^2 = x^2 – 1$ is

(A) $4x^2 + 4y^2 – 3 = 0$ (B) $2x^2 + 2y^2 – 5 = 0$

(C) $3x^2 + 3y^2 – 4 = 0$ (D) none of these

76. If $\left(1-\frac{r_1}{r_2}\right)\left(1-\frac{r_1}{r_3}\right)=2$, where r_1, r_2, r_3 are the ex radii, then the triangle is

(A) equilateral (B) isosceles

(C) right-angled (D) none of these

77. Volume of tetrahedron formed by the planes x + y = 0, y + z = 0, z + x = 0, x + y + z = 1 is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) none of these

Space for rough work

78. The position vectors of three points are $2\vec{a}-\vec{b}+3\vec{c}, \vec{a}-2\vec{b}+\lambda\vec{c}$ & $\mu\vec{a}-5\vec{b}$ where $\vec{a}, \vec{b}$ & $\vec{c}$ are non-coplanar vectors. The points are collinear when

(A) $\lambda=-2, \mu=\frac{9}{4}$ (B) $\lambda=\frac{9}{4}, \mu=2$

(C) $\lambda=\frac{9}{4}, \mu=-2$ (D) none of these

79. The value of the definite integral $\int_{e}^{e^{2010}} \frac{1}{x}\left(1+\frac{1-\ln x}{\ln x.\ln\left(\frac{x}{\ln x}\right)}\right)dx$, equals

(A) $2009-\ln\left((2010)-\ln 2010\right)$ (B) $2010-\ln\left((2009)-\ln 2009\right)$

(C) $2009-\ln\left((2010)-\ln 2009\right)$ (D) $2010-\ln\left((2010)-\ln 2010\right)$

80. Let the equation of the plane containing the line x – y – z – 4 = 0, x + y + 2z – 4 = 0 & is parallel to the line of intersection of the planes 2x + 3y + z = 1 & x + 3y + 2z = 2 be x + Ay + Bz + C = 0 then value of |A + B + C| is

(A) 9 (B) 10

(C) 4 (D) 11

81. Equation of the image of the line $x+y=\sin^{-1}\left(a^6+1\right)+\cos^{-1}\left(a^4+1\right)-\tan^{-1}\left(a^2+1\right), a\in R$ about x-axis is given by

(A) x – y = 0 (B) $x-y=\frac{\pi}{2}$

(C) x – y = π (D) $x-y=\frac{\pi}{4}$

82. If the sum of roots of the equation $\cos 4x-6=7\cos 2x$ in the interval [0, 314] is kπ, k $\in$ R, then k =

(A) 4930 (B) 4940

(C) 4950 (D) None of these

Space for rough work

83. Let $g(x) = f(x)\sin x$, where f(x) is a twice differentiable function on (–∞, ∞), such that $f'(-\pi) = 1$. The value of $g''(-\pi)$ equals

(A) 1 (B) 2
(C) –2 (D) 0

84. The line 3x + 6y = k intersect the curve $2x^2 + 3y^2 + 2xy = 1$ at points A & B. The circle on AB as diameter passes through the origin. The possible value of k is

(A) 3 (B) 4
(C) – 4 (D) 5

85. Let a function f defined from R → R as $f(x) = \begin{cases} x + m, & x < 1 \\ 2mx - 1, & x \geq 1 \end{cases}$. If the function is surjective on R then m must lie in

(A) $[-1, 0)$ (B) [–2, –1]
(C) {2} (D) (0, 2]

86. A man wants to divide 101 coins, a rupee each, among his 3 sons with the condition that no one receives more money than the combined total of other two. The number of ways of doing this is:

(A) ${}^{100}C_2 - 2.{}^{52}C_2$ (B) $\frac{100_{C_2}}{3}$

(C) 1275 (D) $\frac{103_{C_2}}{6}$

87. If $Z_n = \cos\frac{\pi}{n(n+1)(n+2)} + i\sin\frac{\pi}{n(n+1)(n+2)}, n \in N$. then the value of $\lim_{k\to\infty}(z_1 . z_2 . z_3 ..... z_k)$ is.

(A) $\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$ (B) $\frac{-1}{2} + i\frac{\sqrt{3}}{2}$

(C) $\frac{-1}{2} - i\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$

Space for rough work

88. F: R $\to$R be a differentiable function V x$\in$R. if tangent drawn to the curve at any point $\forall$ x $\in$ (a, b) always lie below the curve then

(A) $f'(x) > 0, f''(x) < 0 \forall x \in (a,b)$ (B) $f'(x) < 0, f''(x) < 0 \forall x \in (a,b)$

(C) $f'(x) > 0, f''(x) > 0 \forall x \in (a,b)$ (D) None of these

89. If $x^2y^3 = (x+y)^n, \frac{dy}{dx} = \frac{y}{x}$ then n =

(A) 2 (B) 3

(C) 5 (D) 6

90. Number of integral value of a such that $x^2 + ax + a + 1 = 0$ has integral root is

(A) 1 (B) 2

(C) 3 (D) more than 3

Space for rough work